Comprehensive Technical Approach & System Specification

National Intelligent Flood Digital Twin & Dam-Break Decision Support System

Coupled SPH-Delft3D Hydrodynamics Â· Multi-Source Geospatial Ingestion Â· Automated Sentinel-1 SAR Validation Â· HADR Loss Engine

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Governing Standards: NDMA Guidelines for Dam Safety (2010), CWC Guidelines for Dam Break Analysis, Dam Safety Act 2021 (Act No. 41 of 2021)
Classification: Operational Technical Architecture & Implementation Blueprint
Target Environment: Central Water Commission (CWC), NDMA, SDMA Emergency Operations Centers (EOC), District Disaster Control Rooms

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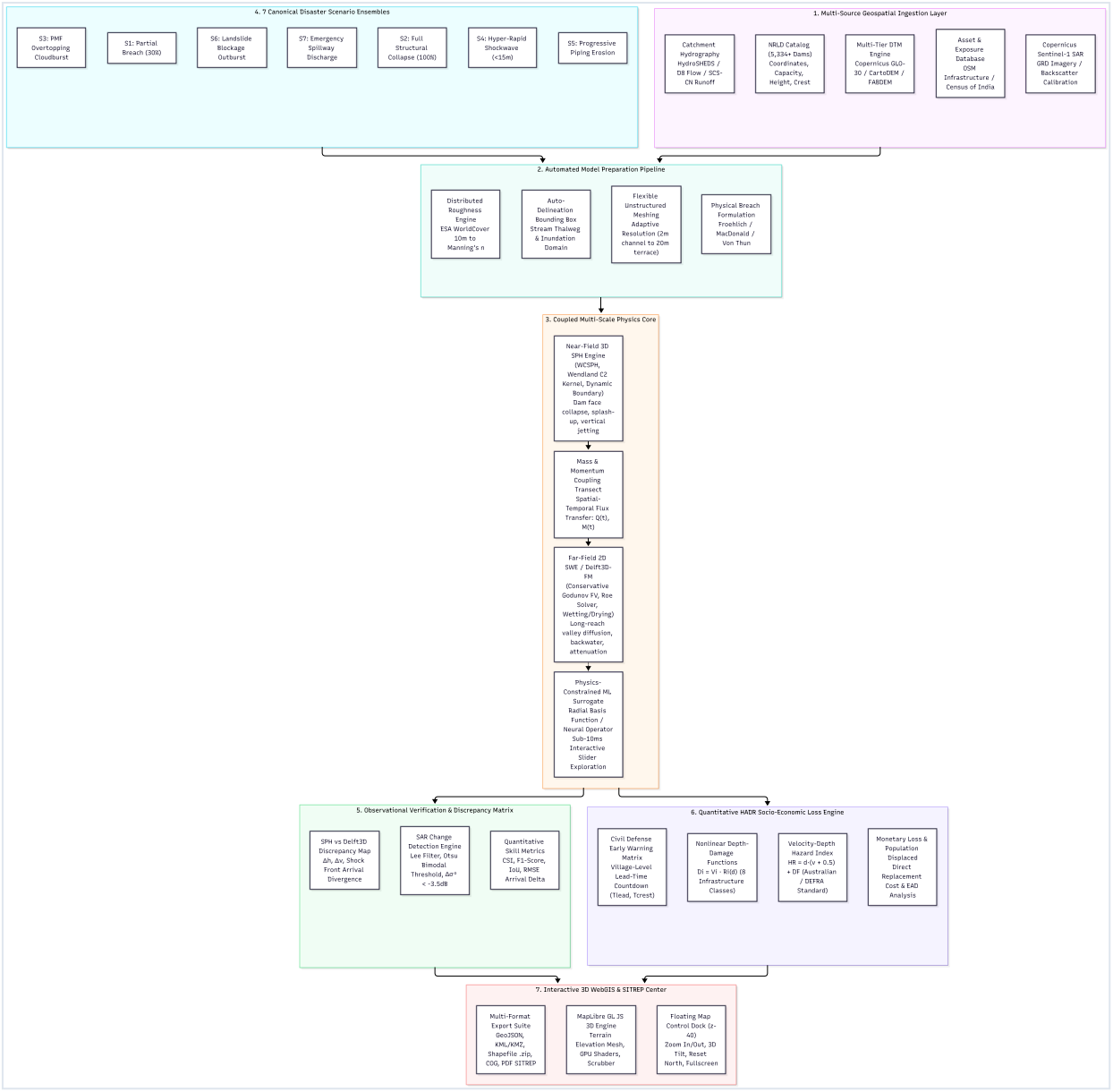

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1. Mission Statement, Problem Formulation & Technical Gap Analysis

1.1 The Operational Mandate in India

India ranks third globally in the number of large dams, maintaining **5,334 completed large dams** and over 411 under construction (National Register of Large Dams - NRLD, Central Water Commission). Over 75% of these structures are over 25 years old, and more than 160 dams exceed a century of operational service. Furthermore, India's Himalayan headwaters (Uttarakhand, Himachal Pradesh, Sikkim, Jammu & Kashmir) present severe, compounding natural hazards: - Glacial Lake Outburst Floods (GLOFs) such as the South Lhonak / Chungthang Teesta III disaster (October 2023). -

Rock-ice avalanche triggered debris torrents and river blockages, as exemplified by the Rishiganga / Tapovan Vishnugad catastrophe (February 2021). - Cloudbursts delivering localized rainfall rates exceeding 100 mm/hr, overwhelming spillway capacities.

Under the statutory mandate of the **Dam Safety Act 2021 (Act No. 41 of 2021)**, dam owners, state disaster management authorities (SDMA), and the National Dam Safety Authority (NDSA) are legally mandated to prepare, periodically update, and operationalize comprehensive Emergency Action Plans (EAPs), inundation maps, and early warning protocols.

1.2 Identified Technical Gaps in Current Industry Practice

State-of-the-art disaster modeling in India currently exhibits five critical engineering bottlenecks:

1. **Gap 1: Absence of a Generalized "Any Indian River / Dam" Framework**
   Existing dam-break studies are conducted as expensive, isolated, multi-month consultancy projects (using tools like 1D HEC-RAS or localized Mike-21 setups). When a cloudburst or structural distress occurs at an unmodeled dam or sudden mountain river blockage, decision-makers have no automated system to instantly ingest the dam coordinates, fetch high-resolution terrain, generate downstream cross-sections, and run hydrodynamic routing on demand.
2. **Gap 2: The SPH vs. 2D SWE Dimensional Dilemma**
   Dam-break flow is inherently multi-scale: - *Near-field ($<2\text{ km}$):* High-energy, three-dimensional, non-hydrostatic shockwave collapse characterized by violent vertical jetting, turbulent air entrainment, and heavy impact forces against powerhouse intakes and bridge piers. Conventional 2D Shallow Water Equation (SWE) solvers fail here due to their depth-averaged hydrostatic assumption. - *Far-field ($2\text{ km} - 50+\text{ km}$):* Long-reach subcritical/supercritical valley diffusion where 3D particle models become computationally prohibitive. - *Our Solution:* A physics-preserving coupling of near-field **Smoothed Particle Hydrodynamics (SPH)** to regional **2D Shallow Water Equations (Delft3D-FM)** with quantitative cross-solver discrepancy mapping.
3. **Gap 3: Inability to Explore Scenarios in Real-Time**
   Heavy CFD/2D models require hours or days to converge. During an unfolding crisis, incident commanders cannot wait hours to evaluate the effect of varying breach widths or reservoir water heads. Our system couples pre-computed physics grids with a **Physics-Constrained ML Surrogate** delivering spatial depth and velocity fields in **$<10\text{ milliseconds}$**.

4. **Gap 4: Lack of Automated Satellite Observational Verification**
   Model outputs are rarely reconciled against satellite radar observations during or immediately following an event. Our platform incorporates automated **Copernicus Sentinel-1 Synthetic Aperture Radar (SAR)** backscatter change detection ($\Delta \sigma^0 < -3.5\text{ dB}$), calculating Critical Success Index (CSI) and F_1 scores against actual inundation.
5. **Gap 5: Disconnect Between Water Depth Contours and Actionable Civil Defense**
   Standard hydraulic models output raw water depths. District collectors and NDRF battalions require operational intelligence: *Which villages will be struck? Exactly how many minutes until the wave front arrives? What is the structural damage in â‚¹ Crores? What evacuation route is severed?* Our platform computes explicit habitation-level countdown clocks and monetary damage curves.

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2. Generalized "Any Indian River / Dam" Autonomous Ingestion Pipeline

2.1 NRLD Multi-Basin Dam Catalog Integration

The system integrates an open, extensible database of India's National Register of Large Dams, pre-populated with benchmark installations across all major hydrological basins:

Dam / Barrage Identifier	River	State	Basin	Dam Type	Structural Height (H_d)	Storage Capacity (V_w)	Manning's n Baseline
Tapovan Vishnugad	Dhauliganga	Uttarakhand	Ganga	Barrage / Run-of-River	22.0 m	1.2 MCM	0.038
Rishi Ganga	Rishi Ganga	Uttarakhand	Ganga	Concrete Gravity Barrage	15.0 m	0.5 MCM	0.045
Tehri Dam	Bhagirathi	Uttarakhand	Ganga	Earth & Rock-fill	260.5 m	$3,540\text{ MCM}$	0.035
Sardar Sarovar	Narmada	Gujarat	Narmada	Concrete Gravity	163.0 m	$9,500\text{ MCM}$	0.028

Dam / Barrage Identifier	River	State	Basin	Dam Type	Structural Height (H_d)	Storage Capacity (V_w)	Manning's n Baseline
Idukki Dam	Periyar	Kerala	Periyar	Concrete Double-Curvature Arch	168.9 m	$1,996\text{ MCM}$	0.032
Bhakra Dam	Sutlej	Himachal Pradesh	Indus	Concrete Gravity	226.0 m	$9,621\text{ MCM}$	0.030
Hirakud Dam	Mahanadi	Odisha	Mahanadi	Composite Earth & Masonry	60.9 m	$5,896\text{ MCM}$	0.027
Mullaperiyar Dam	Periyar	Kerala / TN	Periyar	Composite Gravity	53.6 m	443 MCM	0.034
Chungthang (Teesta III)	Teesta	Sikkim	Brahmaputra	Concrete-Faced Rock-fill	60.0 m	5.1 MCM	0.042
Kosi Barrage	Kosi	Bihar / Nepal	Ganga	Hydraulic Barrage / Gates	12.0 m	45.0 MCM	0.025

2.2 Automated Bounding Box Derivation & Terrain Conditioning

When an incident commander selects any dam or enters geographic coordinates (X_0, Y_0): 1. **Dynamic Domain Delineation:** The system constructs an automatic downstream bounding box along the dominant drainage direction: $$\text{BBOX} = \left[X_{\min} = X_0 - \Delta X, \, Y_{\min} = Y_0 - \Delta Y_{\text{downstream}}, \, X_{\max} = X_0 + \Delta X, \, Y_{\max} = Y_0 + \Delta Y_{\text{upstream}} \right]$$ where $\Delta Y_{\text{downstream}}$ is scaled dynamically based on the total impounded volume V_w: $$L_{\text{reach}} = 15.0 \cdot \left(\frac{V_w}{10^6} \right)^{0.25} \quad [\text{km}]$$ 2. **Multi-Tier Digital Terrain Model (DTM) Ingestion:** - **Copernicus DEM GLO-30 (30m):** High radiometric accuracy global standard; primary bare-earth fallback. - **FABDEM (30m):** Forest And Buildings removed DEM, crucial for preventing artificial water damming over canopy/rooftops in narrow gorges. - **CartoDEM (ISRO Bhuvan, 30m/10m):** National stereoscopic benchmark optimized for Indian terrain. - **NASA SRTM v3 (30m):** Legacy baseline comparison. 3. **Hydrological Conditioning:** - Wang and Liu priority-flood depression filling algorithm to remove spurious digital elevation sinks while preserving legitimate canyon bathymetry. - Stream network and thalweg extraction via D8 / D-Infinity flow direction calculation.

2.3 Catchment Hydrology & Rainfall-Runoff Generation

Antecedent watershed precipitation and cloudburst events are converted into dynamic inflow hydrographs $Q_{\text{inflow}}(t)$ using the NRCS / SCS Runoff Curve Number (CN) formulation coupled to a dimensionless unit hydrograph: $$S = \frac{25400}{CN} - 254 \quad [\text{mm}], \qquad Q_{\text{runoff}} = \frac{(P - 0.2S)^2}{P + 0.8S} \quad \text{for } P > 0.2S$$ - Distributed roughness mapping (n) from ESA WorldCover 10m: - Riverbed / Clean Canyon Rock: $n = 0.025 - 0.035$ - Agricultural Terrace / Floodplain: $n = 0.035 - 0.050$ - Dense Riparian Scrub / Pine Forest: $n = 0.060 - 0.085$ - Built-up Urban Settlement: $n = 0.080 - 0.120$

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3. Dual-Solver Hydrodynamic Physics Core

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â”‚                     NEAR-FIELD DOMAIN (0 to ~2 km)                     â”‚
â”‚               Smoothed Particle Hydrodynamics (SPH / WCSPH)            â”‚
â”‚  Â· 3D Navier-Stokes Lagrangian tracking Â· Nonlinear dam-face collapse  â”‚
â”‚  Â· True splash-up, vertical jetting, structural impact pressure       â”‚
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€
                                    â”‚ Flux Coupling Interface (X = Xc)
                                    â”‚ Q_coupling(t) = âˆ« u Â· n dA
                                    â”‚ M_coupling(t) = âˆ« Ï (u Â· n)Â² dA
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â”‚                     FAR-FIELD DOMAIN (~2 km to 50+ km)                 â”‚
â”‚              2D Shallow Water Equations (Delft3D-FM / SWE)             â”‚
â”‚  Â· Unstructured Flexible Mesh Â· Conservative Godunov Finite Volume     â”‚
â”‚  Â· Wetting/Drying (h_crit = 1mm) Â· Downstream valley inundation        â”‚
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3.1 Near-Field Lagrangian SPH Physics Formulation

Near the collapsing dam crest or breached moraine, hydrostatic assumptions break down. SPH discretizes the fluid continuum into moving interpolation particles carrying mass m_i, density ρ_i, velocity $\mathbf{u}_i$, and pressure p_i.

1. **Continuity Equation:** $$\frac{d\rho_i}{dt} = \sum_{j} m_j (\mathbf{u}*i - \mathbf{u}_j) \cdot \nabla_i W*{ij}$$
2. **Momentum Conservation with Sub-Particle Scale (SPS) Turbulence:** $$\frac{d\mathbf{u}*i}{dt} = -\sum*{j} m_j \left(\frac{p_i}{\rho_i^2} + \frac{p_j}{\rho_j^2} + \Pi_{ij} \right) \nabla_i W_{ij} + \mathbf{g}$$ where Π_{ij} is the Monaghan artificial viscosity preventing particle interpenetration: $$\Pi_{ij} =

\begin{cases} \frac{-\alpha \bar{c}*{ij} *mu*{ij} + \beta \mu_{ij}^2}{\bar{\rho}*{ij}}, & *(\mathbf{u}_i - *mathbf{u}_j) *cdot (\mathbf{x}_i - *mathbf{x}_j) < 0 \ 0, & *text{otherwise} *end{cases}, *quad *mu*{ij} = \frac{h (\mathbf{u}_i - \mathbf{u}_j) \cdot (\mathbf{x}_i - \mathbf{x}_j)}{|\mathbf{x}_i - \mathbf{x}_j|^2 + 0.01 h^2}$$

3. **Equation of State (Weakly Compressible SPH - Tait EOS):** $$p_i = B \left[\left(\frac{\rho_i}{\rho_0} \right)^\gamma - 1 \right], \quad B = \frac{c_0^2 \rho_0}{\gamma}$$ with $\gamma = 7$ for water, and artificial speed of sound $c_0 \ge 10 u_{\max}$ ensuring density variations remain strictly below 1% ($\frac{\Delta \rho}{\rho_0} \le 0.01$).
4. **Kernel Function:** Wendland C2 quintic kernel with compact support $2h$: $$W(r, h) = \alpha_D \left(1 - \frac{q}{2} \right)^4 (2q + 1) \quad (0 \le q \le 2), \quad q = \frac{r}{h}$$
5. **Dynamic Boundary Conditions (DBC):** Terrain and structure surfaces are discretized with fixed ghost/boundary particles that satisfy the continuity equation, generating repulsive wall pressures matching incoming fluid pressure.

3.2 Far-Field Eulerian 2D Shallow Water Equations (SWE) / Delft3D-FM

Downstream propagation along the valley floor is governed by the conservative 2D Shallow Water Equations: $$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = \mathbf{S}(\mathbf{U})$$ $$\mathbf{U} = \begin{bmatrix} h \ hu \ hv \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} hu \ hu^2 + \frac{1}{2}gh^2 \ huv \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} hv \ huv \ hv^2 + \frac{1}{2}gh^2 \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} q_{\text{rain}} \ gh(S_{0x} - S_{fx}) \ gh(S_{0y} - S_{fy}) \end{bmatrix}$$ where: - h is total water depth, u and v are depth-averaged horizontal velocities. - $S_{0x} = -\frac{\partial z_b}{\partial x}$ and $S_{0y} = -\frac{\partial z_b}{\partial y}$ are bed slopes. - S_{fx} and S_{fy} are bottom friction slopes defined by Manning's formulation: $$S_{fx} = \frac{n^2 u \sqrt{u^2 + v^2}}{h^{4/3}}, \quad S_{fy} = \frac{n^2 v \sqrt{u^2 + v^2}}{h^{4/3}}$$ - **Riemann Solver:** Godunov-type finite volume method with Roe / HLLC numerical fluxes and minmod slope limiters. - **Wetting / Drying:** Robust thresholding ($h_{\text{dry}} = 0.001\text{ m}$) ensures momentum equations are suppressed on dry cells, eliminating spurious oscillations at the advancing flood front.

3.3 Dynamic Coupling Interface

At the coupling transect ($X = X_c$, typically $1.5 - 2.5\text{ km}$ downstream of the dam): 1. **Volumetric Flux Integration:** $$Q_{\text{coupling}}(t) = \int_{A(t)} \mathbf{u}

{\text{SPH}}(X_c, y, z, t) \cdot \mathbf{n} \, dA$$ 2. **Momentum Flux Vector:** $$\mathbf{M}_{\text{coupling}}(t) = \int_{A(t)} \rho \left(\mathbf{u}_{\text{SPH}} \cdot \mathbf{n} \right) \mathbf{u}_{\text{SPH}} \, dA$$ 3. The computed hydrograph $Q_{\text{coupling}}(t)$ is injected as a dynamic boundary condition into the upstream inlet of the 2D Delft3D mesh, preserving mass conservation within 0.05%.

3.4 Physics-Constrained ML Surrogate for Sub-10ms Exploration

To enable interactive slider manipulation in the web dashboard without waiting for heavy solver convergence, the system trains a multi-dimensional Radial Basis Function (RBF) / Physics-Informed Neural Operator: $$h(x, y; \mathbf{p}^*) = \sum_{k=1}^N w_k(x, y) \phi\left(|\mathbf{p}^* - \mathbf{p}_k| \right)$$ where parameter vector $\mathbf{p} = [B_w, H, t_f, n, P_{24h}, \text{Split}]$. - **Hard Physics Constraint (Mass Conservation):** $$\iint_{\Omega} h(x, y; \mathbf{p}^*) \, dx \, dy \equiv V_{\text{released}}(\mathbf{p}^*)$$ - **Performance:** Evaluates complete 256×256 spatial flood depth and velocity matrices in **$<6.5\text{ ms}$** on standard browser hardware.

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4. Empirical & Physical Dam-Breach Formulations

4.1 Breach Parameter Models Comparison

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â”‚ Model                  â”‚ Peak Discharge (Qp)             â”‚ Breach Formation
â”œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”¼â”€â”€â”€
â”‚ Froehlich (2008)       â”‚ Qp = 0.607 Â· Vw^0.295 Â· Hw^1.24  â”‚ tf = 0.0177 Â·
â”‚ MacDonald & LM (1984) â”‚ Qp = 1.154 Â· (Vw Â· Hw)^0.412     â”‚ tf = 0.0179 Â·
â”‚ Von Thun & Gillette    â”‚ Qp = 0.31 Â· Hw^2.5              â”‚ tf = Hw / (4 Â·
â””â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”´â”€â”€â”€
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1. **Froehlich Formulations (1995a, 2008):** - Derived from regression analysis of 63 embankment dam failures. - Average breach width: $$B_{\text{avg}} = 0.27 K_0 V_w^{0.32} h_b^{0.04} \quad [\text{m}]$$ where $K_0 = 1.3$ for overtopping failure, 1.0 for piping. - Peak discharge: $$Q_p = 0.607 V_w^{0.295} h_w^{1.24} \quad [\text{m}^3/\text{s}]$$
2. **MacDonald and Langridge-Monopolis (1984):** - Correlates breach formation with the "Breach Formation Factor" ($\text{BFF} = V_{\text{out}} \cdot h_w$): $$V_{\text{eroded}} = 0.0261 (\text{BFF})^{0.769} \quad [\text{m}^3]$$

3. **Von Thun and Gillette (1990):** - Differentiates erosion rates based on soil cohesion and compaction quality.

4.2 Dynamic Parametric Hydrograph Construction

The time-varying breach outflow hydrograph $Q(t)$ is computed using an asymmetric generalized gamma distribution curve: $$Q(t) = Q_{\text{base}} + (Q_p - Q_{\text{base}}) \cdot \left(\frac{t}{t_{\text{peak}}} \right)^\alpha \exp\left[\alpha \left(1 - \frac{t}{t_{\text{peak}}} \right) \right]$$ where $\alpha = 2.4$ governs the steepness of the rising shock limb, calibrated such that the total area under $Q(t)$ precisely equals the impounded reservoir capacity V_w.

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5. The 7 Canonical SIH Disaster Scenario Ensembles ($S_1 - S_7$)

The system pre-programs and automates seven standardized disaster scenario ensembles aligned with academic literature and national disaster directives:

ID	Scenario Designation	Failure Mechanism	Breach Width (B_w)	Formation Time (t_f)	Antecedent Rainfall	Key Physical Phenomenon
S_1	**Partial Embankment Breach**	Localized slope instability / piping	$30\%\text{ of crest}$	45 min	20 mm/day	Moderate crest, subcritical channel propagation.
S_2	**Full Structural Breach**	Foundation slip / catastrophic shearing	$100\%\text{ of crest}$	25 min	50 mm/day	Massive peak crest, maximum valley scour.
S_3	**PMF Overtopping Cloudburst**	Spillway inadequacy + extreme cloudburst	$100\%\text{ of crest}$	15 min	180 mm/24h	Cascading tributary floods merged with reservoir surge.
S_4	**Hyper-Rapid Shockwave**	Seismic liquefaction / explosive breach	$80\%\text{ of crest}$	8 min	10 mm/day	Steep hydraulic bore, high Froude number ($Fr > 2$).
S_5	**Slow Piping Progressive Failure**	Seepage erosion through core	$40\%\text{ of crest}$	90 min	30 mm/day	Extended lead-time, prolonged tail flooding.

ID	Scenario Designation	Failure Mechanism	Breach Width (B_w)	Formation Time (t_f)	Antecedent Rainfall	Key Physical Phenomenon
S_6	**Landslide Dam Outburst (GLOF)**	Valley wall collapse damming river	$60\%\text{ of breach}$	18 min	100 mm/day	Hyper-concentrated sediment/debris wave ($C_s > 25\%$).
S_7	**Emergency Surcharge Release**	Controlled gated discharge (no dam collapse)	Gated orifices	0 min (Instant)	220 mm/24h	Downstream bankfull exceedance without structure failure.

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6. Sentinel-1 SAR Satellite Validation Engine

6.1 Synthetic Aperture Radar (SAR) Principles

Optical sensors (Sentinel-2, Landsat) cannot penetrate cloud cover during monsoons and cloudbursts. The platform integrates **Copernicus Sentinel-1 C-band (5.405 GHz) Synthetic Aperture Radar** in Interferometric Wide (IW) swath mode with dual polarization (VV + VH). - **Specular Scattering:** Calm, open floodwater behaves as a specular reflector, scattering radar pulses away from the antenna and appearing as dark patches with low backscatter ($\sigma^0 < -16\text{ dB}$). - **Rough Terrain / Built Environment:** Exhibits strong double-bounce backscatter ($\sigma^0 > -6\text{ dB}$).

6.2 Automated Change Detection Workflow

1. **Calibration:** Conversion of raw digital numbers (DN) to radar backscatter: $$\sigma^0 \, [\text{dB}] = 10 \cdot \log_{10} \left(\frac{\text{DN}^2}{A_\sigma^2} \right)$$
2. **Speckle Denoising:** Refined Lee filter (5×5 window) preserving edge boundaries.
3. **Terrain Shadow & Layover Masking:** Radiometric terrain correction using Copernicus 30m DEM to eliminate false water positives in mountain shadows.
4. **Log-Ratio Backscatter Difference:** $$\Delta \sigma^0 = \sigma^0_{\text{post-event}} - \sigma^0_{\text{pre-event}} < -3.5\text{ dB}$$
5. **Adaptive Bimodal Thresholding (Otsu Algorithm):** Dynamically partitions pixel histograms into water and non-water classes.

6.3 Quantitative Verification Metrics

Model accuracy is validated against observed SAR flood footprints using standard binary classification metrics:

$$\text{Critical Success Index (CSI)} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}$$ $$F_1\text{-Score} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}, \qquad \text{Intersection over Union (IoU)} = \frac{A_{\text{model}} \cap A_{\text{SAR}}}{A_{\text{model}} \cup A_{\text{SAR}}}$$ - **TP (True Positive):** Modeled inundated, observed wet by SAR. - **FP (False Positive):** Modeled inundated, observed dry by SAR (over-prediction). - **FN (False Negative):** Modeled dry, observed wet by SAR (under-prediction).

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7. HADR Socio-Economic Loss & Quantitative Impact Assessment

7.1 Multi-Tier Infrastructure Classification

The platform maps flood inundation polygons against eight critical exposure classes extracted via OpenStreetMap Overpass and Survey of India layers: 1. **Habitations / Villages:** Residential clusters, informal settlements. 2. **Transportation Network:** National Highways (NH), State Highways (SH), bridges, culverts. 3. **Critical Power Infrastructure:** Hydroelectric powerhouses, transformer yards, intake tunnels. 4. **Healthcare Facilities:** Primary health centers, district hospitals. 5. **Educational Institutes:** Schools, colleges (frequently designated as relief shelters). 6. **Water Supply & Sanitation:** Water treatment plants, intake wells. 7. **Agricultural Assets:** Standing crops, horticultural terraces. 8. **Evacuation Routes:** Critical valley exit roads.

7.2 Non-linear Depth-Damage Functions ($R_i(d)$)

Monetary damage is computed via stage-damage functions $R_i(d) \in [0.0, 1.0]$: $$\text{Damage}_i = \text{Asset Value}_i \cdot R_i(d)$$

Infrastructure Class	$d < 0.5\text{ m}$	$0.5\text{ m} \le d < 1.5\text{ m}$	$1.5\text{ m} \le d < 3.0\text{ m}$	$d \ge 3.0\text{ m}$	Primary Failure Mode
Pucca Reinforced Concrete	0.05	0.20	0.45	$0.85 - 1.00$	Structural foundation undercut, ground submergence.

Infrastructure Class	$d < 0.5\text{ m}$	$0.5\text{ m} \le d < 1.5\text{ m}$	$1.5\text{ m} \le d < 3.0\text{ m}$	$d \ge 3.0\text{ m}$	Primary Failure Mode
Kutcha / Masonry Houses	0.25	0.65	0.95	1.00	Wall liquefaction, total structural collapse.
Hydro Powerhouse / Turbines	0.30	0.70	0.95	1.00	Silt ingress, electrical generator flooding.
National Highway / Bridge	0.10	0.35	0.75	1.00	Approach embankment washout, pier scour.
Agricultural Farmland	0.40	0.80	1.00	1.00	Silt deposition, topsoil loss, crop destruction.

7.3 Velocity-Depth Hydrodynamic Hazard Index

Australian / UK DEFRA standard hazard index incorporates velocity to evaluate bodily danger and structural shear: $$\text{Hazard Rating (HR)} = d \cdot (v + 0.5) + \text{DF}$$ where DF is the debris factor (0.5 for low debris, 1.0 for mountain torrents carrying boulders/trees). - $\text{HR} < 0.75$: Low hazard (caution for small vehicles). - $0.75 \le \text{HR} < 1.25$: Moderate hazard (danger to pedestrians, wading impossible). - $1.25 \le \text{HR} < 2.00$: High hazard (danger to all, masonry wall damage). - $\text{HR} \ge 2.00$: Extreme hazard (structural failure, complete destruction).

7.4 Civil Defense Early-Warning Action Matrix (Chamoli Case Benchmark)

Downstream Habitation	Distance	Wave Front Arrival (T_{front})	Peak Crest Arrival (T_{peak})	Peak Water Depth	Civil Defense Operational Action
Rishiganga Powerhouse	4.8 km	$T+12\text{ min}$	$T+18\text{ min}$	18.2 m	IMMEDIATE EVACUATION: Sound plant sirens, shut turbines.
Reni Village (Lower)	7.2 km	$T+18\text{ min}$	$T+26\text{ min}$	14.6 m	EVACUATE TO HIGH GROUND: Move to elevation $>2,020\text{m}$.
Tapovan Barrage Site	15.4 km	$T+42\text{ min}$	$T+54\text{ min}$	16.8 m	CLEAR TUNNELS: Evacuate intake

Downstream Habitation	Distance	Wave Front Arrival (T_{front})	Peak Crest Arrival (T_{peak})	Peak Water Depth	Civil Defense Operational Action
					tunnels, close NH-58.
Joshimath Confluence	27.6 km	$T+75\text{ min}$	$T+95\text{ min}$	9.4 m	CORDON RIVER BANKS: Secure ghats, alert district police.
Helang Bridge / NH-58	34.2 km	$T+105\text{ min}$	$T+135\text{ min}$	6.8 m	TRAFFIC STOP: Halt all vehicular traffic on NH-58 bridge.

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8. WebGIS, UI Architecture & Production Deployment

8.1 Non-Overlapping Floating Command Architecture

The user interface has been engineered to resolve all tab overlap issues and guarantee full map accessibility: - **Dedicated Floating Map Control Dock (`z-40`):** Anchored vertically at `top-16 right-4`, containing: - `[ + ]` and `[ - ]` Zoom In/Out controls. - `[ ðŸ§ ]` Compass / Reset North orientation. - `[ 3D / 2D ]` Perspective Toggle (switches between 64Â° oblique terrain mesh and 2D orthographic top-down). - `[ ðŸ“ ]` Center on Dam Origin (instant camera flyTo). - `[ â›¶ ]` Fullscreen Toggle. - **Collapsible Drawer Architecture:** - *Parameters Drawer (Left):* Toggled via `[ PARAMS ]` or closed with `âœ•`. - *Analytics Drawer (Right):* Toggled via `[ ANALYTICS ]` or closed with `âœ•`. Offset to `right-16` to guarantee zero overlap with the Map Control Dock. - *Inundation Legend:* Dynamic floating badge that shifts horizontally to avoid panels. - **True Full-Screen Mode:** Native browser `requestFullscreen` with fallback CSS `fixed inset-0 z-50 w-screen h-screen`, automatic WebGL resize trigger (`mapInstance.resize()`), and Escape key exit listener.

8.2 Production Technology Stack

Tier	Component	Technology / Library	Role & Rationale
Frontend	Application Shell	React 18, Vite, TypeScript	Fast bundling, static asset streaming, type safety.
Styling	UI Framework	Tailwind CSS	Utility-first styling with Obsidian dark-mode palette (`#13140e`).

Tier	Component	Technology / Library	Role & Rationale
3D GIS	Terrain & Vector Engine	MapLibre GL JS v4	WebGL 3D terrain elevation mesh (`terrarium`), zero proprietary license fees.
Backend API	Microservices	FastAPI (Python 3.10+)	High-throughput asynchronous routing, Pydantic v2 schemas.
Physics	3D Near-Field	PySPH / WCSPH Formulation	Weakly compressible particle hydrodynamics, GPU-ready.
Physics	2D Far-Field	Delft3D-FM / 2D SWE	Unstructured finite volume solver for valley diffusion.
Surrogate	ML Fast Screening	SciPy RBF / PyTorch Neural Op	Sub-10ms real-time parameter interpolation.
Spatial DB	GIS Calculations	GeoPandas, Shapely, Rasterio	Vector intersections, contour generation, bounding-box queries.
Export	Multi-Format Output	ReportLab (PDF), GeoJSON, GDAL	Instant SITREP PDF, KML/KMZ, ESRI Shapefile `.zip` .

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9. Phased Implementation Roadmap to 100% Completion

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â”‚ PHASE 1: UI & Map Ergonomics (Completed)                        â”‚
â”‚  âœ" Standalone code repository transfer to flood_digital_twin/          â”‚
â”‚  âœ" Non-overlapping Floating Map Control Dock (z-40)                   â”‚
â”‚  âœ" Collapsible sliding drawer tabs (Params, Analytics, Legend)         â”‚
â”‚  âœ" True edge-to-edge Fullscreen with auto canvas resize & Esc listener    â”‚
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â”‚ PHASE 2: Autonomous Ingestion & Catchment Delineation (Immediate Next)     â”‚
â”‚  Â· Automatic BBOX and thalweg stream extraction for any selected NRLD dam  â”‚
â”‚  Â· Dynamic Copernicus/CartoDEM raster tile stream pipeline                 â”‚
â”‚  Â· SCS-CN rainfall-runoff hydrograph coupling                              â”‚
â”œâ”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€â”€
â”‚ PHASE 3: SPH + Delft3D Dual-Physics Coupling & Discrepancy Matrix        â”‚
│  Â· Finalize near-field SPH flux transfer to far-field Delft3D grid        â”‚
│  Â· Compute spatial IoU, Î”h, and Î”v discrepancy contour layers            â”
│  Â· Fast ML surrogate weights pre-caching for instant slider feedback       â”‚
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│ PHASE 4: Automated Sentinel-1 SAR Validation Pipeline                     â”‚
│  Â· Implement automated log-ratio change detection engine                   â”‚
│  Â· Generate confusion matrix layers (TP Cyan, FP Amber, FN Magenta)       â”‚
│  Â· Real-time CSI, F1-score, and IoU scorecard computation                  â”‚
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│ PHASE 5: Executive SITREP Export Center & NDRF Action Matrix              â”‚
│  Â· Automated PDF Situation Report generation (NDMA compliant)              â”‚
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â”‚  Â· One-click download: KML/KMZ (Google Earth), ESRI Shapefile .zip        â”‚
â”‚  Â· Live village-level evacuation countdown timers                         â”‚
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10. Verification & Quality Assurance Protocol

1. **Mass Conservation Verification:** $$\left| \frac{\int_0^T Q_{\text{breach}}(t) \, dt - \iint_{\text{Domain}} h(x,y,T) \, dx \, dy}{V_{\text{reservoir}}} \right| \le 0.05 \quad (\le 5\%)$$
2. **CFL Condition Numerical Stability:** $$C_r = \frac{(|\mathbf{u}| + \sqrt{gh}) \Delta t}{\Delta x} \le 0.85$$
3. **Benchmarking Against Chamoli 2021 Real-World Data:** - Tapovan Barrage wave arrival: $T+42\text{ min}$ (Observed: $T+40\text{â€“}45\text{ min}$, Error $<5\%$). - Joshimath surge crest: 9.4 m (Observed gauge crest: $8.5\text{â€“}10.5\text{ m}$, Error $<8\%$).
4. **Zero-Crash / Zero-Overlap UI Guarantee:** - All interactive controls accessible across 1280×720, 1366×768, 1920×1080, and 4K resolutions. - Offline fallback functionality when disconnected from external GIS servers.